

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
int i= 1; int s =1; while(s <= n)
{
i++;
s += i;
}
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

```
#include <stdio.h>
```

```
int main()
{
int n;
```

```
int i = 1;
int s = 1;
int c = 3;
scanf("%d",&n);
```

```
while (s <= n)
{
c++;
i++;
c++;
s+=i;
c++;
}
```

```
printf("%d", c);
```

```
return 0;
```

```
}
```

2. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
```

```
{
```

```
if(n==1)
```

```
{
```

```
printf("*");
```

```
}
```

```
else
```

```
{
```

```
for(int i=1; i<=n; i++)
```

```
{
```

```
for(int j=1; j<=n; j++)
```

```
{
```

```
printf("*");
```

```
printf("*");
```

```
break;
```

```
}
```

```
}
```

```
}
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n;
```

```
scanf("%d", &n);
```

```
int count = 1;
```

```
if (n == 1)
```

```
{
```

```
count++;
```

```
}
```

```
else
```

```
{
```

```
count++;
```

```
for (int i = 1; i <= n; i++)
```

```
{
```

```
count++;
```

```
count++;
```

```
for (int j = 1; j <= n; j++)
```

```
{
```

```
count++;
```

```
count++;
```

```
count++;
```

```
break;
```

```
}
```

```
}
```

```
}
```

```
printf("%d", count);
```

```
return 0;
```

```
}
```

3. Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
```

```
{
```

```
for (i = 1; i <= num; ++i)
```

```

{
if (num % i== 0)
{
printf("%d ", i);
}
}
}

```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int num,i,counter=0;
```

```
scanf("%d",&num);
```

```
for ( i = 1; i <= num; i++)
```

```
{
```

```
counter++;
```

```
if (num % i == 0)
```

```
{
```

```
counter++;
```

```
}
```

```
}
```

```
counter +=num+1;
```

```
printf("%d", counter);
```

```
return 0;  
}
```